CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CE/IT) Examination

April 2013
CE201 Data Structure and Algorithms

Date: 16.04.2013, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) Give the difference between data type and data structure. Also mention types of data structure with an example. **[04]**

(b) What is searching? Differentiate Linear Search and Binary Search. **[03]**

Q - 2 (a) Write an algorithm or C program to implement push and pop operations of stack using an array. **[05]**

(b) Translate following infix expression into postfix expression using a stack. **[04]**

(A^B) / D * E / (F +A) *(D * E) - C

(c) Explain different types of file indexing methods stating their advantages and disadvantages. **[05]**

OR

(c) Answer the following questions: **[05]**
(i) Difference between array and linked list.
(ii) Write recursive steps to solve Tower of Hanoi problem. Which data structure is used for Tower of Hanoi problem?

Q - 3 (a) Write an algorithm for singly linked list that performs following operations: **[06]**
(i) Insert a node at the end of the linked list
(ii) Delete a node with information field X

(b) Define queue. Write C functions for inserting an element into simple queue. **[04]**

OR

Q - 3 (a) Write at algorithm for doubly linked list that performs following operations: **[06]**
(i) Insert a node after the node whose address is M
(ii) Delete the node whose address is OLD

(b) Write a short note on priority queue. **[04]**

Q - 3 (c) Trace the following data using Insertion Sort. **[04]**

5, 9, 2, 15, 30, 92, 1, 24

SECTION – II

Q - 4 (a) Trace the following data using Bubble Sort. **[04]**

19, 85, 12, 10, 8, 78, 35

(b) Define following terms: **[03]**

(i) path (ii) cycle (iii)isolated node

Q - 5 (a) Write a short note on threaded binary tree. **[05]**

OR

(a) Create binary search tree for following value **[05]**

15, 25, 39, 5, 10, 832, 34, 7, 6, 28

(b) Explain circular queue. How it differs from simple queue? **[03]**

(c) Explain quick sort using recursive algorithm. **[06]**

OR

(c) Explain merge sort with detailed algorithm. **[06]**

Q - 6 (a) Explain various multiple key access file organization in brief. **[05]**

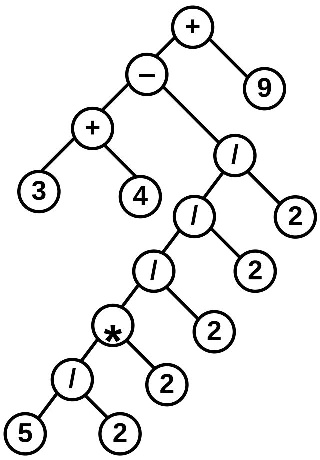
(b) Define graph. Explain storage representation of a graph. **[05]**

OR

Q - 6 (a) Explain different methods used for defining hash functions. **[05]**

(b) What is Minimum Heap Tree? Perform ascending order sorting for following data using Minimum Heap Tree : 15, 19, 10, 7, 17, 16 **[05]**

Q - 6 (c) Traverse the following tree into post order and pre order. Evaluate the equation step by step using stack. **[04]**



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